

Laporan Mini Project 2

News & HR Analytics API (FastAPI dan PostgreSQL)

Nama	Yuono Dwi Raharjo
Tanggal	27 September 2026
Repository	https://github.com/YuonoD020602/digital-skola-mini-project-data-pipeline
Folder proyek	Mini Project 2_data-pipeline

1. Ringkasan Proyek

Proyek ini melanjutkan pipeline data dari sesi sebelumnya. Data artikel hasil scraping (526 artikel) dan data attrition karyawan (1.470 baris) yang sudah tersimpan di PostgreSQL disajikan melalui REST API berbasis FastAPI. Akses ke endpoint dilindungi dengan API key, dan tersedia alur autentikasi JWT Bearer token.

2. Endpoint yang Dibangun

Task	Endpoint	Fungsi
1	GET /articles/search	Pencarian artikel dengan filter (sumber, tanggal) dan pagination.
2	GET /articles/stats	Statistik jumlah artikel secara total dan per sumber.
3	GET /attrition/department/{dept_name}	Statistik attrition, pendapatan, dan top earners per departemen; 404 jika departemen tidak ada.
4	GET /attrition/risk-profile	Daftar karyawan berisiko tinggi berdasarkan kriteria OverTime, masa kerja, dan pendapatan.
Tambahan	POST /token	Menerbitkan JWT access token (berlaku 30 menit) dari client credentials.

3. Ringkasan Hasil Pengujian (Postman)

No	Skenario	Request	Status
SS1	Pencarian artikel tanpa filter	GET /articles/search	200 OK
SS2	Pencarian dengan filter sumber dan tanggal	GET /articles/search?source=bbc&date_from=2026-09-20	200 OK
SS3	Pagination halaman 2	GET /articles/search?page=2	200 OK
SS4	Statistik artikel	GET /articles/stats	200 OK
SS5	Detail attrition departemen Sales	GET /attrition/department/Sales	200 OK
SS6	Profil risiko karyawan (API key)	GET /attrition/risk-profile	200 OK
SS7a	Permintaan JWT token	POST /token	200 OK
SS7b	Profil risiko dengan JWT Bearer	GET /attrition/risk-profile?limit=5	200 OK
Bonus	Penanganan departemen yang tidak ada	GET /attrition/department/Finance	404 Not Found

Seluruh pengujian dijalankan di Postman terhadap server lokal (<http://127.0.0.1:8000>) menggunakan collection `MiniProject2.postman_collection.json` yang juga tersedia di repository.

4. Dokumentasi Screenshot Postman

SS1. Pencarian artikel tanpa filter

Request: GET /articles/search | Status: 200 OK

Mini Project 2 — News & HR > ... > SS1. Search - tanpa filter (default pagination)

GET `{{base_url}}/articles/search` Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body `JSON` Preview Visualize

```
88 {
89   "source": "quotes.toscrape",
90   "content": "\"The truth.\" Dumbledore sighed. \"It is a beautiful and
91     terrible thing, and should therefore be treated with great caution.\"
92     -- J.K. Rowling. Tags: truth.",
93   "published_at": "2026-09-25T02:01:40.358453",
94   "scraped_at": "2026-09-25T04:01:40.518488"
95 },
96 {
97   "page": 1,
98   "per_page": 10,
99   "total_items": 526,
100  "total_pages": 53
101 }
```

Pagination default: page 1, 10 artikel per halaman. Total 526 artikel dalam 53 halaman.

SS2. Pencarian dengan filter sumber dan tanggal

Request: GET /articles/search?source=bbc&date_from=2026-09-20 | Status: 200 OK

Mini Project 2 — News & HR A... > ... > SS2. Search - filter source + date_from

GET `{{base_url}}/articles/search?source=bbc&date_from=2026-09-20` Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
☒ source	bbc			
☒ date_from	2026-09-20			
Key	Value	Description		

Body `JSON` Preview Visualize

```
88 {
89   "source": "bbc",
90   "content": "\"The Israeli delegation could be be seen cheering in support
91     during Netanyahu's speech.",
92   "published_at": "2026-09-24T18:29:29",
93   "scraped_at": "2026-09-25T04:01:40.518488"
94 },
95 {
96   "page": 1,
97   "per_page": 10,
98   "total_items": 45,
99   "total_pages": 5
100 }
```

Filter bekerja: hasil turun dari 526 menjadi 45 artikel (5 halaman), seluruhnya bersumber BBC dan terbit sejak 20 September 2026.

SS3. Pagination halaman 2

Request: GET /articles/search?page=2 | Status: 200 OK

Mini Project 2 — News & HR Analytics API > 6. Mini Pr... > SS3. Search - page 2

GET `{{base_url}}/articles/search?page=2` Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
page	2			
Key	Value	Description		

Body `200 OK` • 576 ms • 5.02 KB • Save Response

```
{
  "source": "quotes.toscrape.com/#quote-01-ALBERT-EINSTEIN",
  "content": "\"Anyone who has never made a mistake has never tried anything new.\" -- Albert Einstein. Tags: mistakes.",
  "published_at": "2026-09-25T02:01:37.734675",
  "scraped_at": "2026-09-25T04:01:40.518488"
},
{
  "page": 2,
  "per_page": 10,
  "total_items": 526,
  "total_pages": 53
}
```

Menampilkan halaman ke-2 dari 53 halaman; total 526 artikel sama dengan SS1 karena tidak memakai filter.

SS4. Statistik artikel

Request: GET /articles/stats | Status: 200 OK

Mini Project 2 — News & HR Analytics API > 6. Mini Projec... > SS4. Article stats

GET `{{base_url}}/articles/stats` Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body `200 OK` • 508 ms • 281 B • Save Response

```
{
  "total_articles": 526,
  "by_source": {
    "books.toscrape": 200,
    "quotes.toscrape": 100,
    "nytimes": 81,
    "bbc": 52,
    "cnn": 40,
    "aljazeera": 25,
    "techcrunch": 20,
    "skynews": 8
  }
}
```

Total 526 artikel. Rincian per sumber: books.toscrape 200, quotes.toscrape 100, nytimes 81, bbc 52, cnn 40, aljazeera 25, techcrunch 20, skynews 8 (jumlahnya sesuai total).

SS5. Detail attrition departemen Sales

Request: GET /attrition/department/Sales | Status: 200 OK

Mini Project 2 — News & HR Analytics ... > ... > SS5. Department detail - Sales

GET `{{base_url}}/attrition/department/Sales` Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body `JSON` Preview Visualize

200 OK • 250 ms • 489 B • Save Response

```
1 {
2   "department": "Sales",
3   "total_employees": 446,
4   "attrition_count": 92,
5   "attrition_rate": 0.2063,
6   "avg_income": 6959.17,
7   "min_income": 1052,
8   "max_income": 19847,
9   "top_earners": [
10    {
11      "EmployeeNumber": 1282,
12      "JobRole": "Manager",
```

446 karyawan, 92 attrition (attrition rate 0,2063 atau 20,63%). Rata-rata pendapatan 6.959,17 (min 1.052, maks 19.847), disertai daftar top earners.

SS6. Profil risiko karyawan (API key)

Request: GET /attrition/risk-profile | Status: 200 OK

Mini Project 2 — News & HR Analytics API > 6. Mini Project ... > SS6. Risk profile

GET `{{base_url}}/attrition/risk-profile` Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body `JSON` Preview Visualize

200 OK • 556 ms • 3.61 KB • Save Response

```
1 {
2   "criteria": {
3     "OverTime": "Yes",
4     "YearsAtCompany": "< 3",
5     "MonthlyIncome": "< rata-rata departemen"
6   },
7   "total_high_risk": 82,
8   "returned": 20,
9   "data": [
10    {
11      "EmployeeNumber": 701,
12      "Department": "Research & Development",
13      "JobRole": "Research Scientist",
```

Kriteria risiko tinggi: OverTime = Yes, YearsAtCompany < 3, dan MonthlyIncome di bawah rata-rata departemen. Ditemukan 82 karyawan; 20 data pertama ditampilkan.

SS7a. Permintaan JWT token

Request: POST /token | Status: 200 OK

The screenshot shows a REST client interface with a sidebar on the left containing a list of collections. The main panel displays a POST request to `POST {{base_url}} /token` with a status of **200 OK**. The response body is a JSON object:

```
{  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJ3dW9uby1jbG11bnQpYXQzE30TAzMjQ1NjksImV4cCI6MTc5MDMyNjM2OSwiaWZpdjoiZTUwYmZmZDZjODRiMjY3OTVhY2RmZDZjODAxM2QyNTkiQ3F1fDZ2tttAadFmEkSSW1_LU57i1DJ8RiR0AzivEgw",  "token_type": "bearer",  "expires_in": 1800}
```

Server menerbitkan `access_token` berformat JWT dengan `token_type` bearer dan masa berlaku 1.800 detik (30 menit).

SS7b. Profil risiko dengan JWT Bearer

Request: GET /attrition/risk-profile?limit=5 | Status: 200 OK

The screenshot shows a REST client interface with a sidebar on the left containing a list of collections. The main panel displays a GET request to `GET {{base_url}} /attrition/risk-profile?limit=5` with a status of **200 OK**. The response body is a JSON object:

```
{  "criteria": {    "OverTime": "Yes",    "YearsAtCompany": "< 3",    "MonthlyIncome": "< rata-rata department"  },  "total_high_risk": 82,  "returned": 5,  "data": [    {      "EmployeeNumber": 701,      "Department": "Research & Development",      "JobRole": "Research Scientist",      "YearsAtCompany": 1,      "MonthlyIncome": 1009,      "dept_avg_income": 6281.25,      "Attrition": "Yes"    }  ]}
```

Endpoint diakses memakai JWT Bearer token dari SS7a. Parameter `limit` bekerja: 5 data ditampilkan dari total 82 karyawan berisiko tinggi.

Bonus. Penanganan departemen yang tidak ada

Request: GET /attrition/department/Finance | Status: **404 Not Found**

The screenshot shows a REST client interface with a sidebar on the left containing a list of collections. The main panel displays a GET request to `{{base_url}}/attrition/department/Finance`. The response status is **404 Not Found** with a response time of 229 ms and a body size of 183 B. The response body is a JSON object:

```
{  "detail": "Department 'Finance' tidak ditemukan"}
```

The interface also includes tabs for Docs, Params, Authorization, Headers, Body, Scripts, and Settings. A 'Query Params' table is visible, and a 'Body' tab shows the JSON response. A 'Fix Department Not Found Error' button is present in the response area.

Error handling berjalan: server tidak crash, melainkan membalas 404 dengan pesan “Department 'Finance' tidak ditemukan”. Dataset hanya memiliki departemen Sales, Research & Development, dan Human Resources.

5. Catatan Keamanan

File `.env` yang berisi kredensial database, API key, dan client secret tidak diunggah ke repository. Repository juga dilengkapi `.gitignore` untuk mengecualikan `.env`, `__pycache__`, dan file `*.pyc`.